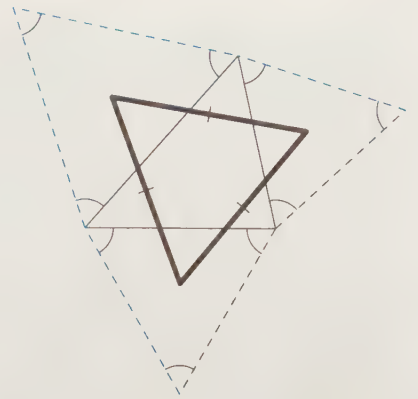




Outer Napoleon Triangle

Equality may perhaps be a right, but no power on earth can ever turn it into a fact. – Honore de Balzac

CHAPTER 3

Congruent Triangles

3.1 Introduction

Two figures are **congruent** if they are exactly the same – in other words, we can slide, spin, and/or flip one figure so that it is exactly on top of the other figure.

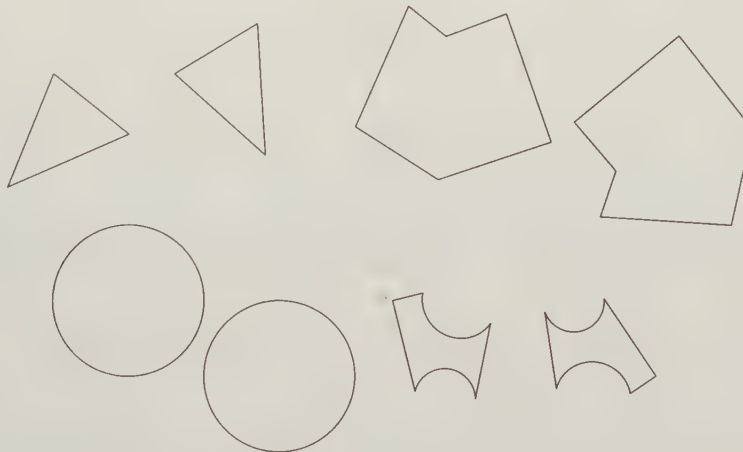


Figure 3.1: Congruent Figures

Figure 3.1 shows four pairs of congruent figures. In each case, we can take one of the figures and slide it, spin it, and/or flip it so that it exactly coincides with the other.

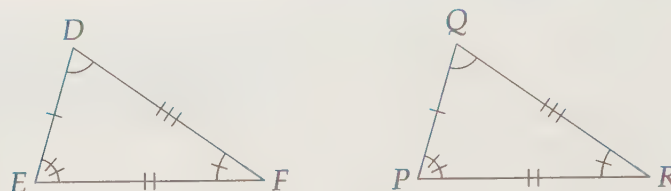


Figure 3.2: Congruent Triangles

We use the symbol $\cong$ to denote that two figures are congruent. For example, to describe the congruent triangles in Figure 3.2, we write $\triangle DEF \cong \triangle QPR$. Notice that we are careful to put the vertices in the same order: D corresponds to Q , E to P , and F to R . We would not, for example, write $\triangle DEF \cong \triangle PQR$ for the triangles in Figure 3.2.

If we have two congruent triangles, all the corresponding pairs of sides are equal, as are the corresponding angles. Conversely, if all pairs of corresponding sides of two triangles have equal lengths, and all the corresponding angles of the two triangles are equal, then the triangles are congruent.

Fortunately, to show that two triangles are congruent, we don't have to go through the hassle of proving each pair of sides and each pair of corresponding angles are congruent. In this chapter, we'll cover various ways requiring considerably less information to show that two triangles are congruent.

3.2 SSS Congruence

Problems

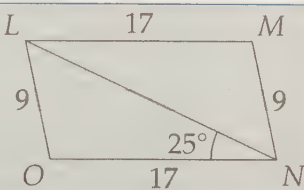
Problem 3.1: In each of the three triangles below, the side lengths are 5, 7, and 8 units. Are the measures of the angles the same in all three triangles? (Yes, you should use your protractor.)



Problem 3.2: In this problem we investigate how many different non-congruent triangles have sides with lengths 2.5 cm, 3.5 cm, and 4 cm by building triangle ABC with these side lengths. Start by drawing segment $\overline{AB}$ with length 4 cm.

- Suppose C is 3.5 cm away from A . What points on your paper are 3.5 cm away from A ? Draw the figure that consists of all these points.
- We have $AB = 4$ cm, and our answer to (a) tells us where C must be so that $AC = 3.5$ cm. We also want C to be 2.5 cm away from B . What points on your paper are 2.5 cm from B ? Draw the figure that consists of all these points.
- Where in your resulting diagram can C possibly be?
- How many different non-congruent triangles have sides with lengths 2.5 cm, 3.5 cm, and 4 cm?

Problem 3.3: In the figure, the sides and angle have measures as shown. Find $\angle MLN$.



Problem 3.4: In this problem we will show that if a radius of a circle bisects (cuts in half) a chord of the circle that is not a diameter, then the radius is perpendicular to the chord.

- Draw a circle with center O and a chord connecting points A and B on the circle.
- Draw a radius of the circle that passes through the midpoint, M , of $\overline{AB}$, thereby bisecting $\overline{AB}$.
- Find a pair of congruent triangles and use these triangles to show that $\angle OMA = \angle OMB = 90^\circ$.

Often our best way to get started with a new geometric principle is to experiment. So, we start off by examining some triangles that have the same side lengths and checking if the triangles are indeed congruent.

Problem 3.1: In each of the three triangles below, the side lengths are 5, 7, and 8 units. Are the measures of the angles the same in all three triangles? (Yes, you should use your protractor.)



Solution for Problem 3.1: Using our protractor, we find that the angles of each triangle are approximately 82° , 60° , and 38° . Each triangle has the largest angle opposite the largest side and the smallest angle opposite the smallest side. Hence, the sides and the angles appear to be the same in all three triangles, so it looks like they're congruent. $\square$

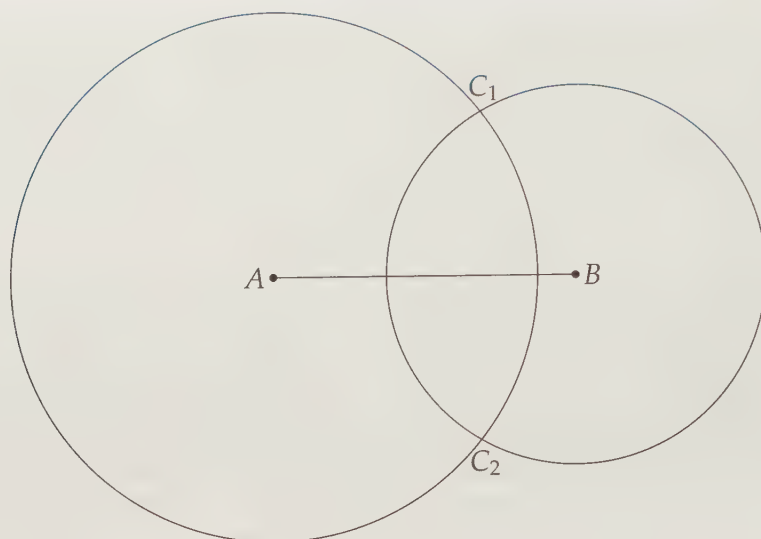
Let's see if we can get a common-sense explanation for why the triangles of Problem 3.1 are congruent.

Problem 3.2: Investigate how many different non-congruent triangles have sides with lengths 2.5 cm, 3.5 cm, and 4 cm by building triangle ABC with these side lengths.

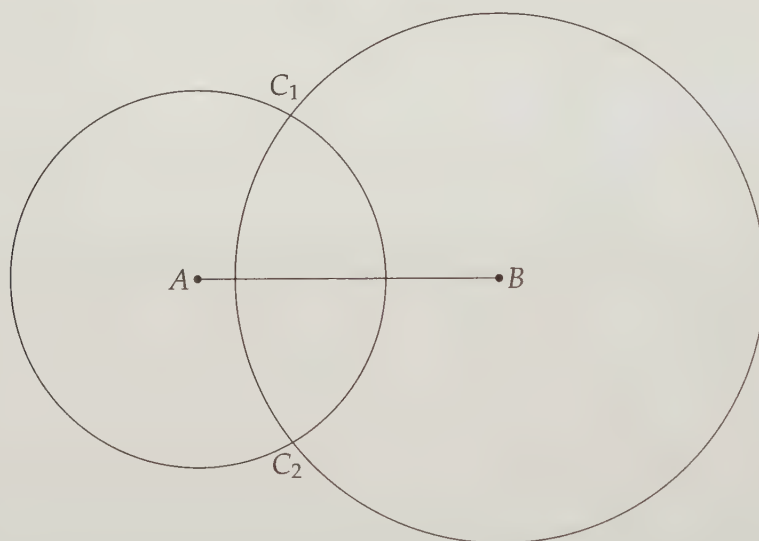
Solution for Problem 3.2: It doesn't matter which side we set to 4 cm, so we just pick $\overline{AB}$ to be the long side. We start building $\triangle ABC$ by drawing a 4 cm long segment.



Now we have to use our other side lengths to figure out where C can possibly be. Suppose $AC = 3.5$ cm. Then, C must be 3.5 cm away from A . So, C must be on a circle with center A and radius 3.5 cm. Similarly, since $BC = 2.5$ cm, C must lie on a circle with center B and radius 2.5 cm. We draw both circles and end up with the diagram below. Since C must be on both of the circles, C must be at one of the intersection points. Our two options for C (labeled C_1 and C_2) are merely mirror images of each other.



If we had instead assumed $AC = 2.5$ cm and $BC = 3.5$ cm, we would end up with essentially the same picture, as shown below.



Notice that if we 'flip' our first diagram horizontally, we get the second one. In fact, we don't really even need this second diagram. We could have simply swapped the labels A and B in our first diagram to investigate the possibility that $AC = 2.5$ cm and $BC = 3.5$ cm. $\square$

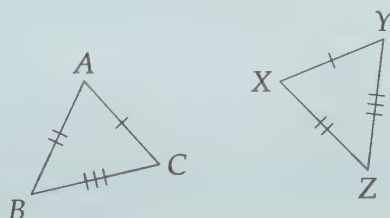
Concept: Recognizing two seemingly different situations as the same can be a powerful simplifying tool in math problems. In geometry, this often involves recognizing **symmetry**, as we did in noting that C_1 and C_2 are mirror images of each other in each diagram above, and in noting that the two diagrams themselves are mirror images of each other.

Important: We call the principle illustrated in Problems 3.1 and 3.2 the **Side-Side-Side Congruence Theorem**, or **SSS Congruence** for short. SSS states:



If the lengths of the sides of one triangle equal the lengths of the corresponding sides of another triangle, then the triangles are congruent.

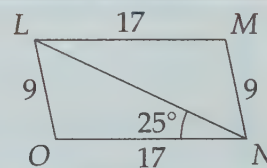
For example, in the diagram below, we have $\triangle ABC \cong \triangle XZY$ by SSS Congruence.



Our 'proof' that all triangles with side lengths 2.5 cm, 3 cm, and 4 cm are congruent skips over a few very important points. First, we didn't prove that triangles $\triangle ABC_1$ and $\triangle ABC_2$ are congruent in each of our cases. Nor did we prove that the triangles in the first case are congruent to the triangles in the second. Intuitively, it seems clear that these triangles are all congruent, but we haven't proved it. In fact, with the tools we have at this point, we cannot prove SSS Congruence at all!

We must accept as an axiom one of the Congruence Theorems we will study in this chapter. We can then use that one theorem, and some other tools, to prove the others. Since some of those other tools are considerably more advanced than those we have now, we'll stick to our intuitive explanations for all of the Congruence Theorems.

Problem 3.3: In the figure, the sides and angle have measures as shown. Find $\angle MLN$.



Solution for Problem 3.3: What's wrong with this:

Bogus Solution: Since $\overline{LM}$ and $\overline{NO}$ are parallel, $\angle MLN = \angle LNO = 25^\circ$.



This Bogus Solution assumes that $\overline{LM} \parallel \overline{NO}$. This might be true, but we have to prove it to use it.

Since $LO = MN$, $ON = LM$, and $LN = LN$, we have $\triangle LNO \cong \triangle NLM$ by SSS. Since angles $\angle MLN$ and $\angle ONL$ are corresponding parts of these two triangles, they must be equal. Therefore,

$$\angle MLN = \angle ONL = 25^\circ.$$

Notice that we can now use $\angle MLN = \angle ONL$ to prove that $\overline{LM} \parallel \overline{NO}$. $\square$

Important:



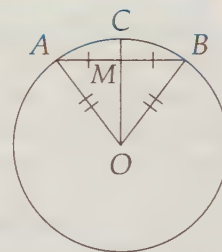
Problem 3.3 shows us a very important general use of congruent triangles. Once we determine that two triangles are congruent, we can apply whatever we know about the sides and angles of one triangle to the other triangle. This obvious principle often goes by the fancy name ‘Corresponding Parts of Congruent Triangles are Congruent,’ or CPCTC.

Unless your teacher tells you to, you don’t have to clutter your paper writing CPCTC all over. Once you have proved that $\triangle LNO \cong \triangle NLM$, you can simply write, ‘Since $\triangle LNO \cong \triangle NLM$, we have $\angle MLN = \angle ONL$.’

Let’s try using congruent triangles in a proof.

Problem 3.4: Prove that if a radius of a circle bisects (cuts in half) a chord of the circle that is not a diameter, then the radius must be perpendicular to the chord.

Solution for Problem 3.4: We start with a diagram, drawing circle O , chord $\overline{AB}$ with midpoint M , and the radius through M . We then mark the equal pieces of the chord and the equal radii of the circle, and see that we have congruent triangles. (Radii is the plural of radius.) Specifically, $\triangle AMO \cong \triangle BMO$ by SSS. So, $\angle AMO = \angle BMO$. Since $\angle AMO + \angle BMO = 180^\circ$, we must have $\angle AMO = \angle BMO = 90^\circ$. $\square$



Important:



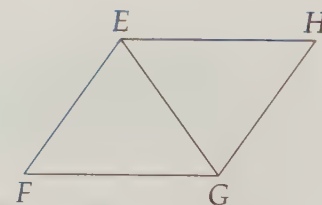
If a radius of a circle bisects a chord of the circle that is not a diameter, then the radius must be perpendicular to the chord.

As we’ll see in Chapter 6, this works in reverse, too! A radius perpendicular to a chord must bisect the chord.

Exercises

3.2.1 $EF = GH$ and $FG = EH$ in the diagram at right.

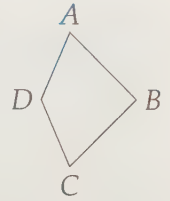
- Prove that $\triangle EFG \cong \triangle GHE$.
- Show that $\angle EGF = \angle GEH$.
- Show that $\overline{HE} \parallel \overline{FG}$.
- Show that $\overline{HG} \parallel \overline{EF}$.



3.2.2 In triangle ABC , $AB = AC$. Let M be the midpoint of side $\overline{BC}$.

- A segment, line, or ray **bisects** an angle if it divides it into two angles with equal measure. Show that $\overline{AM}$ bisects $\angle BAC$ by proving that $\angle BAM = \angle CAM$.
- Show that $\overline{AM} \perp \overline{BC}$.

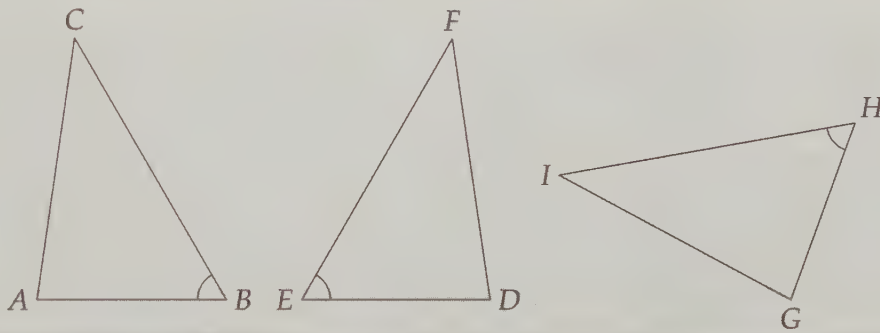
3.2.3 In the diagram at right, $AB = 7$, $AD = 4$, $CD = 4$, and $BC = 7$. Prove that $\angle ABD = \angle CBD$. **Hints:** 165



3.3 SAS Congruence

Problems

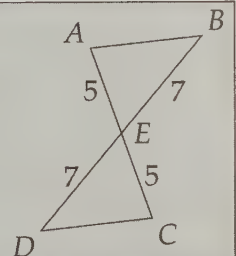
Problem 3.5: In each of the triangles below, two sides have lengths 2.5 cm and 4 cm, and the angle between these sides is 60° . Measure the third side and the other two angles of each triangle.



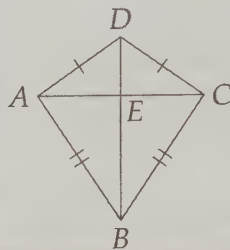
Problem 3.6: Investigate why the triangles in the previous problem appear to be congruent by constructing a triangle with side lengths 2.5 cm and 4 cm such that the angle between these two sides is 60° .

- Draw segment $\overline{AB}$ such that $AB = 4$ cm.
- Let $\overline{BC}$ be the side that is 2.5 cm long; thus, C is 2.5 cm from B . Construct the figure that consists of all points that are 2.5 cm from B .
- The angle between $\overline{AB}$ and $\overline{BC}$ must be 60° . Where can C be to make $\angle ABC = 60^\circ$? (Forget about $BC = 2.5$ cm for this part.)
- Draw your answer to (c).
- Where can C possibly be located?
- Are all triangles with two sides of length 2.5 cm and 4 cm with an angle of 60° between them congruent?

Problem 3.7: Point E is the midpoint of both $\overline{AC}$ and $\overline{BD}$ as shown. Prove that $\overline{AB} \parallel \overline{CD}$.

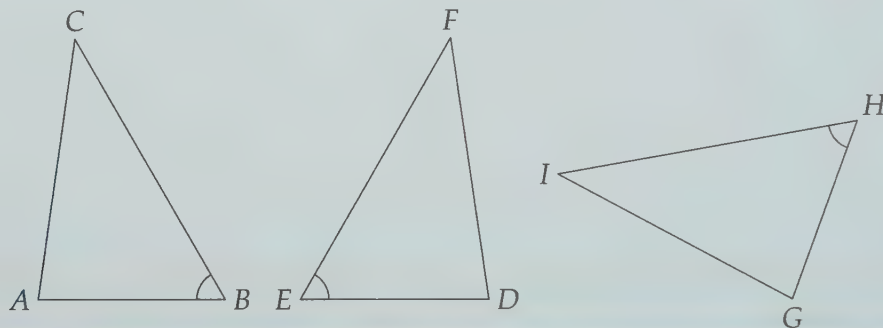


Problem 3.8: In the figure below, $AD = CD$ and $AB = BC$. Prove that $\overline{DB} \perp \overline{AC}$. **Hints:** 461



Showing that all the sides of one triangle equal those of another is not the only way to quickly prove two triangles are congruent. In this section we investigate one way to use angles along with sides.

Problem 3.5: In each of the triangles below, two sides have lengths 2.5 cm and 4 cm, and the angle between these sides is 60° . Measure the third side and the other two angles of each triangle.



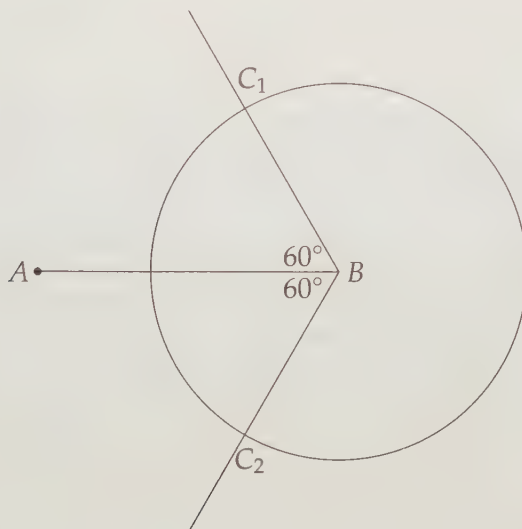
Solution for Problem 3.5: In each case, the third side is 3.5 cm and the other two angles are approximately 82° and 38° , with the 82° angle opposite the 4 cm side. The triangles sure look congruent. $\square$

Sidenote: Perhaps only the ancient Greeks have contributed as much to geometry as the German mathematician **David Hilbert**. Among his wide range of work was *Grundlagen der Geometrie*, which placed geometry in a formal axiomatic framework. Instead of Euclid's 5 axioms, Hilbert used 21 axioms, since Euclid's work included many hidden assumptions that must be accepted as axioms. Among those axioms is SAS Congruence, which, in combination with other axioms, he used to prove the other Congruence Theorems presented in this chapter.

Not only did Hilbert produce much great mathematics, but he also inspired and challenged others. In 1900 he gave an address to the Second International Congress in Paris on 23 unsolved problems of great importance. Since then, some of the problems have been solved, but many still puzzle mathematicians to this day. He also succinctly phrased the passion that guides many great mathematicians and scientists when he closed an address with the words, "We must know, we shall know."

Problem 3.6: Investigate why the triangles in the previous problem appear to be congruent by constructing a triangle with side lengths 2.5 cm and 4 cm such that the angle between these two sides is 60° .

Solution for Problem 3.6: We start off just like we did in Problem 3.2, by letting let $AB = 4$ cm. We want $BC = 2.5$ cm and $\angle ABC = 60^\circ$. Since C is 2.5 cm from B , it lies on a circle with center B and radius 2.5 cm. Since $\angle ABC = 60^\circ$, we know that C lies on a ray from B that forms a 60° angle with $\overline{AB}$. We can use our compass and protractor to build the circle and our two possible rays as shown below.



In our diagram there are only two options for C : the two points where one of the rays meets the circle. Just like we saw in Problem 3.2, these C 's are mirror images. $\triangle ABC_1$ and $\triangle ABC_2$ have the same side lengths and angle measures.

Therefore, we find that no matter how we build the triangle given two sides and the angle between them, we always get a triangle with the same sides and angles. $\square$

Important: We call the principle illustrated in Problem 3.6 the **Side-Angle-Side Congruence Theorem**, or **SAS** for short. SAS states:



If two sides of one triangle and the angle between them are equal to the corresponding sides and angle of another triangle, then the two triangles are congruent.

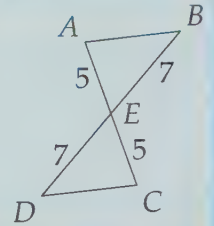
As with our illustration of SSS on page 51, our solution to Problem 3.6 is not a complete proof of the SAS Congruence Theorem. It merely provides an intuitive explanation. Let's see SAS in action.

Extra! *It is impossible to be a mathematician without being a poet in soul.*



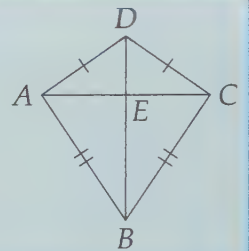
—Sofia Kovalevskaya

Problem 3.7: Point E is the midpoint of both $\overline{AC}$ and $\overline{BD}$ as shown. Prove that $\overline{AB} \parallel \overline{CD}$.



Solution for Problem 3.7: We saw in Section 2.7 that we can use angle equality to prove that lines are parallel. We can get the angle equality we need by using congruent triangles. Since $AE = EC$, $BE = ED$, and $\angle AEB = \angle CED$ (because they are vertical angles), we have $\triangle AEB \cong \triangle CED$ by SAS. Therefore, $\angle ABE = \angle CDE$, so $\angle ABD = \angle CDB$ and we have $\overline{AB} \parallel \overline{CD}$. $\square$

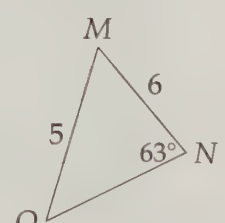
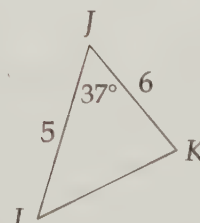
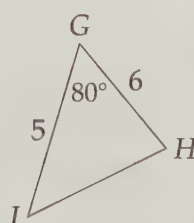
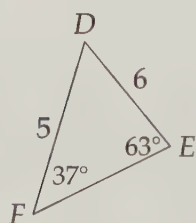
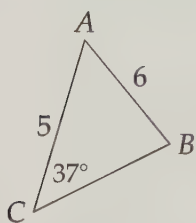
Problem 3.8: In the figure, $AD = CD$ and $AB = BC$. Prove that $\overline{DB} \perp \overline{AC}$.



Solution for Problem 3.8: We have to prove that two lines are perpendicular, but we don't have any information about angles. We look for congruent triangles since that will give us some angle information. The given side equalities along with the obvious $BD = BD$ give us $\triangle DAB \cong \triangle DCB$ by SSS. This doesn't give us any information about angles at point E , but it does give us some angle equalities like $\angle ADE = \angle CDE$. This, together with $AD = DC$ and $DE = DE$, gives us $\triangle ADE \cong \triangle CDE$ by SAS. Therefore, $\angle AED = \angle CED$. Since these two angles are also supplementary, we have $\angle AED = \angle CED = 90^\circ$, so $\overline{BD} \perp \overline{AC}$. $\square$

Exercises

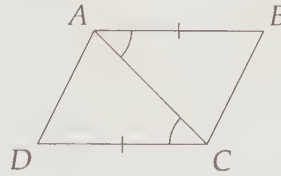
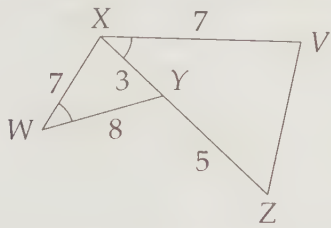
3.3.1 Which two of the triangles below must be congruent and why must they be congruent?



3.3.2 In triangle ABC , $AB = AC$. Let M be the point on $\overline{BC}$ such that $\overline{AM}$ bisects $\angle BAC$ (so that $\angle CAM = \angle BAM$).

- Show that M is the midpoint of $\overline{BC}$.
- Show that $\overline{AM} \perp \overline{BC}$.

3.3.3 Find VZ in the diagram at the left below. (Note: The diagram is not to scale.) **Hints:** 467

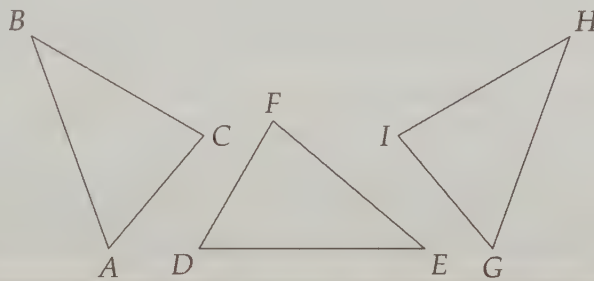


3.3.4 In the figure to the right above, $AB = CD$ and $\angle BAC = \angle DCA$. Prove that $\overline{AD} \parallel \overline{BC}$.

3.4 ASA and AAS Congruence

Problems

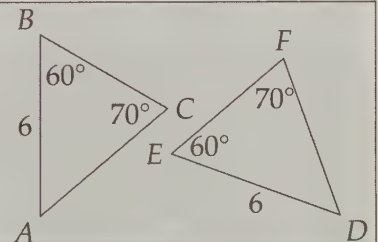
Problem 3.9: Shown are three triangles with two angles measuring 60° and 40° , and each triangle has a side of length 3 cm between these two angles. Measure the other two sides of each triangle. Make a guess about triangle congruence given two angles and an included side.



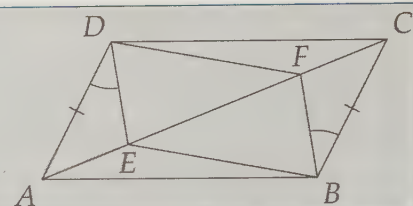
Problem 3.10: Investigate why the triangles in the previous problem are congruent by constructing a triangle with angles 60° and 40° such that the side between the two angles has length 3 cm. Start by drawing a segment $\overline{AB}$ such that $AB = 3$ cm.

- Let the angle between $\overline{AB}$ and $\overline{BC}$ be 60° . Where can C be to make $\angle ABC = 60^\circ$?
- Draw your answer to (a). The angle between $\overline{AB}$ and $\overline{AC}$ must be 40° . Where can C be to make $\angle BAC = 40^\circ$? (Forget about $\angle ABC = 60^\circ$ for this part.)
- Draw your answer to (b). Where can C possibly be located?
- Are all triangles with a side of length 3 cm between angles with measures 60° and 40° congruent?

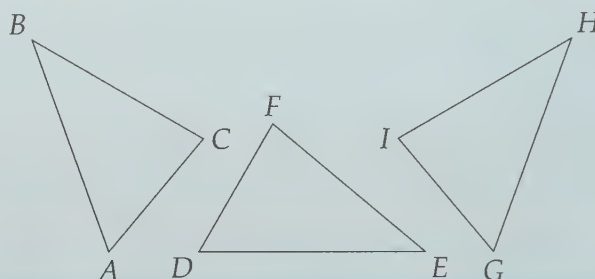
Problem 3.11: What if the equal sides are not between the equal angles? For example, what if $AB = DE = 6$, $\angle B = \angle E = 60^\circ$, and $\angle C = \angle F = 70^\circ$ as shown? Can we conclude that $\triangle ABC \cong \triangle DEF$? Why or why not?



Problem 3.12: In the diagram, $AD = BC$, $\overline{AD} \parallel \overline{BC}$, and E and F are on $\overline{AC}$ so that $\angle ADE = \angle CBF$. Prove that $\overline{AB} \parallel \overline{CD}$ and $DF = EB$.



Problem 3.9: Shown are three triangles with two angles measuring 60° and 40° , and each triangle has a side of length 3 cm between these two angles. Measure the other two sides of each triangle. Make a guess about triangle congruence given two angles and an included side.

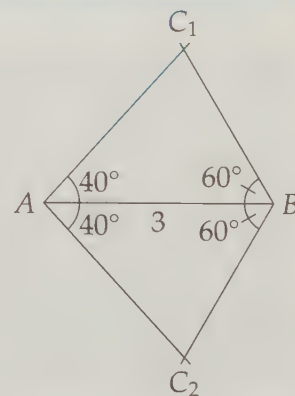


Solution for Problem 3.9: In each triangle, the side opposite the 40° angle has length about 2 cm, and the side opposite the 60° angle has length around 2.6 cm. The triangles appear to be congruent. $\square$

Problem 3.10: Investigate why the triangles in the previous problem are congruent by constructing a triangle with angles 60° and 40° such that the side between the two angles has length 3 cm.

Solution for Problem 3.10: We start with $\overline{AB}$ of length 3 cm. We let $\angle ABC = 60^\circ$. As in Problem 3.6, this means that point C is on one of two rays from B that make an angle of 60° with $\overline{AB}$. Similarly, we have $\angle BAC = 40^\circ$, which means that C is on one of the two rays from A that make an angle of 40° with $\overline{AB}$. Point C must be at one of the two intersections of rays. The two potential C 's, marked C_1 and C_2 , are mirror images of each other.

Therefore, we find that all the triangles we can build given two angles and the side between them are congruent. $\square$



Extra! Divide the figure at right into four congruent pieces that can be rearranged to form a square.

Source: ThinkFun Inc.'s Puzzles webpage. See the links page mentioned in the Acknowledgements for a link to the site.

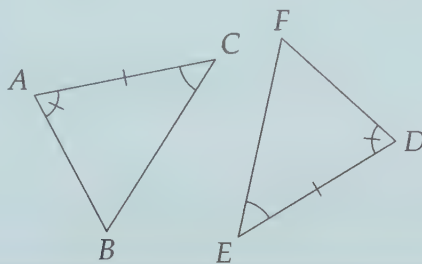


Important:

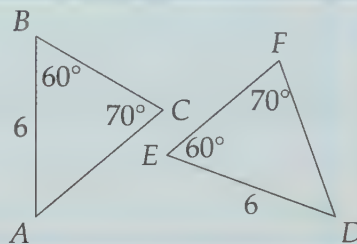
We call the principle illustrated in Problem 3.10 the **Angle-Side-Angle Congruence Theorem**, or **ASA** for short. ASA states that

If two angles of one triangle and the side between them are equal to the corresponding angles and side of another triangle, then the two triangles are congruent.

For example, in the diagram below, we have $\triangle ABC \cong \triangle DFE$.



Problem 3.11: What if the equal sides are not between the equal angles? For example, what if $AB = DE = 6$, $\angle B = \angle E = 60^\circ$, and $\angle C = \angle F = 70^\circ$? Can we conclude that $\triangle ABC \cong \triangle DEF$? Why or why not?



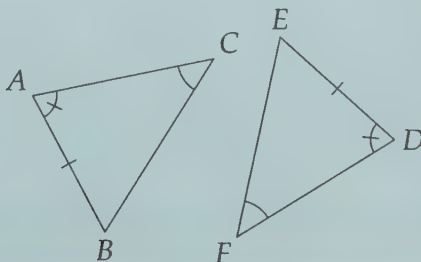
Solution for Problem 3.11: If we have two angles of a triangle, we have the third. In the diagram above we can quickly see that $\angle D = \angle A = 50^\circ$. Hence, we can conclude that $\triangle ABC \cong \triangle DEF$ by ASA. $\square$

When we have the situation illustrated above, we don't have to find the third angle and invoke ASA. This procedure always works, so it gets an unsurprising theorem name of its own.

Important:

We call the principle illustrated in Problem 3.11 the **Angle-Angle-Side Congruence Theorem**, or **AAS** for short. AAS states:

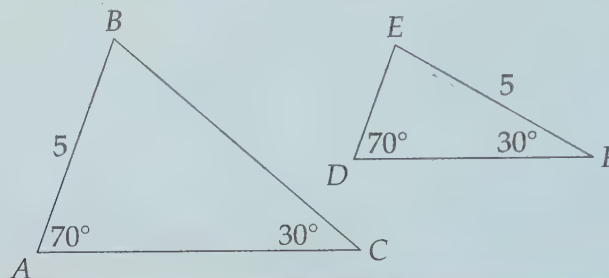
If two angles and a side of one triangle equal the corresponding angles and side in another triangle as shown below, then the triangles are congruent.



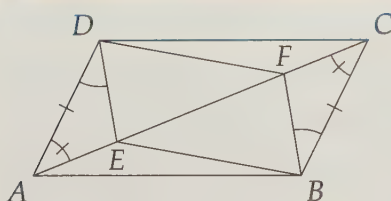
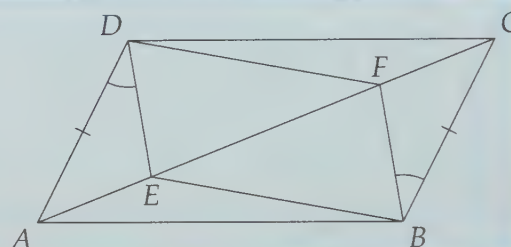
WARNING!!



When using AAS, the equal sides must be adjacent to corresponding equal angles. For example, the triangles shown below are **not** congruent because the equal sides are not adjacent to corresponding equal angles!



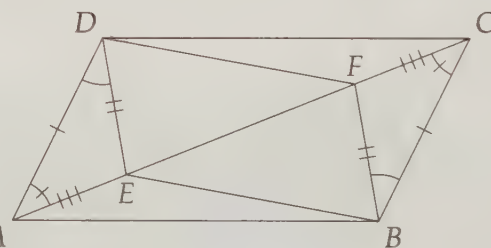
Problem 3.12: In the diagram, $AD = BC$, $\overline{AD} \parallel \overline{BC}$, and E and F are on $\overline{AC}$ so that $\angle ADE = \angle CBF$. Prove that $\overline{AB} \parallel \overline{CD}$ and $DF = BE$.



Solution for Problem 3.12: We go hunting for equal angles to use to prove that $\overline{AB}$ and $\overline{CD}$ are parallel. We start by marking the angles we know are equal. We have a pair in our original diagram, and can add $\angle DAC = \angle BCA$ due to $\overline{AD} \parallel \overline{BC}$, as shown to the left.

Now we can find some congruent triangles. Since $AD = BC$, $\angle DAC = \angle BCA$, and $AC = AC$, we have $\triangle CAD \cong \triangle ACB$ by SAS. Therefore, $\angle DCA = \angle BAC$, so $\overline{AB} \parallel \overline{CD}$.

We also have another pair of congruent triangles: $\triangle DAE \cong \triangle BCF$ by ASA. It's not immediately clear how this will help us prove $BE = DF$, so we go ahead and mark the new equal lengths we know from our new triangle congruence. Focusing on segments $\overline{BE}$ and $\overline{DF}$, we look for congruent triangles that have these as sides. We start by looking for triangles we know something about, which leads us to $\triangle ADF$ and $\triangle CBE$. We already have a side and an angle, and $AF = AE + EF = CF + EF = CE$ gives us a second side. So, $\triangle ADF \cong \triangle CBE$ by SAS. Finally, we conclude that $DF = BE$.



We could also have used SAS to show $\triangle DFE \cong \triangle BEF$, so $DF = BE$. $\square$

Concept:



In complicated geometry problems, mark side and angle equalities as you find them (particularly when you find non-obvious ones!)

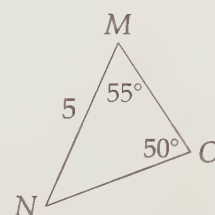
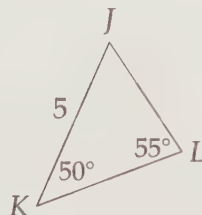
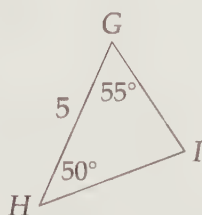
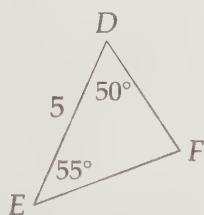
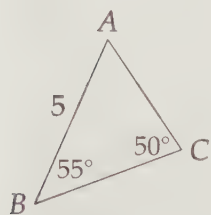
Extra! The real voyage of discovery consists not in seeking new landscapes, but in having new eyes.



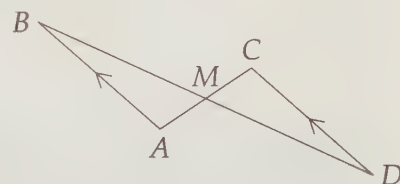
—Marcel Proust

Exercises

3.4.1 Find all pairs of triangles below that must be congruent. Write out the appropriate congruence (make sure you have the vertices in the right order!), and explain why the triangles must be congruent.



3.4.2 In the diagram at right, $\overline{AB} \parallel \overline{DC}$. If M is the midpoint of $\overline{AC}$, must it also be the midpoint of $\overline{BD}$? Why or why not? **Hints:** 183

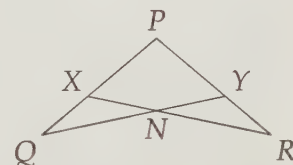


3.4.3 Use ASA Congruence to prove that AAS Congruence is a valid Congruence Theorem. (Do not assume AAS Congruence is a valid theorem for this part – you are asked here to show that any two triangles that satisfy the AAS criteria are indeed congruent *without using AAS*.) **Hints:** 463

3.4.4 In the figure, $PQ = PR$ and $\angle PQY = \angle PRX$.

(a) Prove that $QY = RX$.

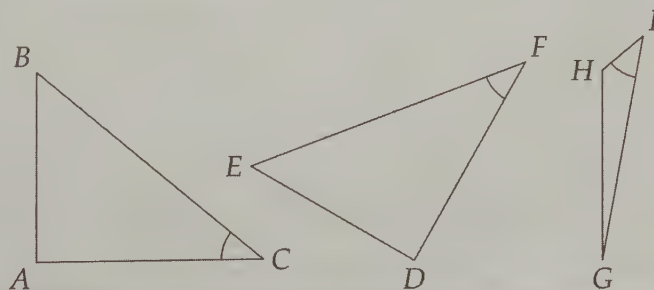
(b)★ Prove that $XN = YN$. **Hints:** 158, 343



3.5 SSA Not-Necessarily Congruence

Problems

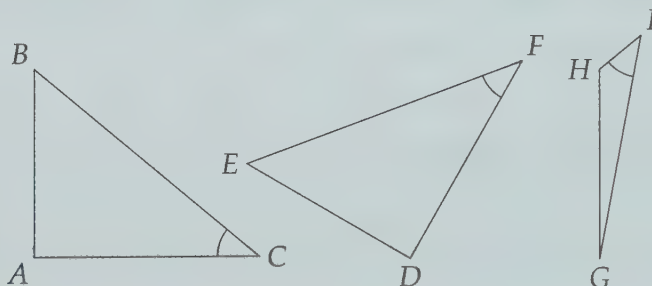
Problem 3.13: In the diagram below, we have three triangles with two sides of length 2.5 cm and 3 cm, and an angle of 40° . The 40° angle has the 3 cm segment as a side, but does not have the 2.5 cm segment as a side. Are the triangles congruent?



Problem 3.14: In this problem we explore why there's no 'Side-Side-Angle Congruence Theorem' by building a triangle with $AB = 1.8$ cm, $BC = 1.5$ cm, and $\angle BAC = 40^\circ$.

- Draw $AB = 1.8$ cm.
- We know that $BC = 1.5$ cm. Draw the points in the diagram that are 1.5 cm from B .
- Draw the points in the diagram that can be C such that $\angle BAC = 40^\circ$. (Forget $BC = 1.5$ cm for now.)
- Use the previous two parts to determine where C can possibly be.
- Is SSA a valid congruence theorem?
- Do we ever have a case in which given the lengths of sides $\overline{AB}$ and $\overline{BC}$, and $\angle BAC$, there's only one possible length of $\overline{AC}$?

Problem 3.13: In the diagram, we have three triangles with two sides of length 2.5 cm and 3 cm, and an angle of 40° . The 40° angle has the 3 cm segment as a side, but does not have the 2.5 cm segment as a side. Are the triangles congruent?



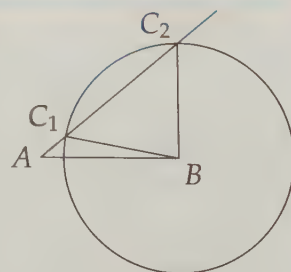
Solution for Problem 3.13: Um, no. Clearly HI is way smaller than EF and BC , so the triangles aren't congruent. $\square$

SSS works. SAS works. ASA, and AAS, too. But SSA, not so much. Why?

Problem 3.14: Figure out why there's no 'Side-Side-Angle Congruence Theorem' by building a triangle with $AB = 1.8$ cm, $BC = 1.5$ cm, and $\angle BAC = 40^\circ$.

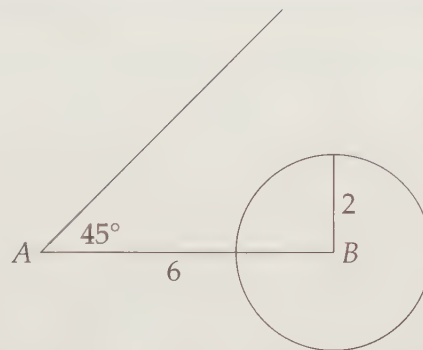
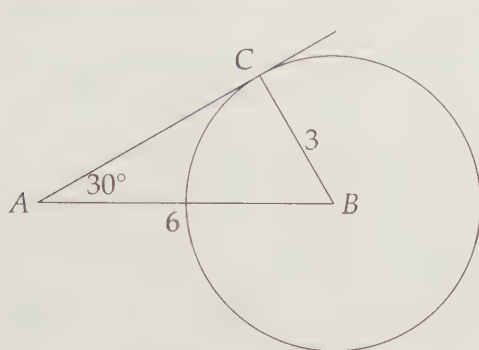
Solution for Problem 3.14: We start as we have before, drawing $\overline{AB}$ with length 1.8 cm. Since $BC = 1.5$ cm, C must be on a circle with center B and radius 1.5 cm. Also, since $\angle BAC = 40^\circ$, C must be on a ray from A that makes a 40° angle with $\overline{AB}$. The ray and the circle are shown to the right.

Uh-oh. The ray hits the circle in two points, C_1 and C_2 . Both $\triangle ABC_1$ and $\triangle ABC_2$ match the information we are given about $\triangle ABC$, but they are very obviously not congruent. $\square$



WARNING!!

Side-Side-Angle (SSA) is **not** a valid congruence theorem. You cannot use it to prove that two triangles are congruent.



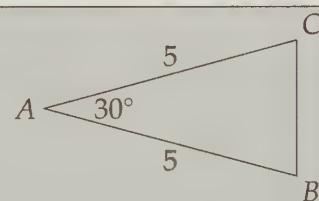
You might have noticed that if we try to build a triangle given two sides and a non-included angle as in Problem 3.14, we don't always get two possible points. The left diagram above shows a case in which we do get exactly one possible C , and the right figure above shows a case in which no triangles are possible. We'll be revisiting the former case later when we have more tools to talk about this special case.

3.6 Isosceles and Equilateral Triangles

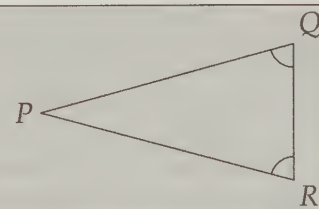
Problems

Problem 3.15: In the diagram, $AB = AC = 5$ and $\angle CAB = 30^\circ$.

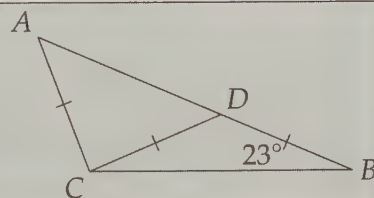
- Let M be the midpoint of $\overline{BC}$. Draw $\overline{AM}$. Prove that $\triangle ACM \cong \triangle ABM$.
- Find $\angle AMB$.
- Find $\angle ACB$.



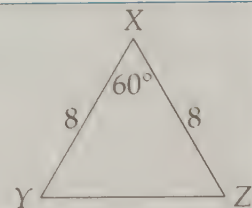
Problem 3.16: Prove that if $\angle PQR = \angle PRQ$, then $PR = PQ$.



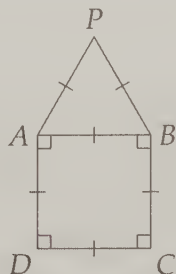
Problem 3.17: In the diagram, $AC = CD = DB$, and $\angle B = 23^\circ$. Find $\angle A$.



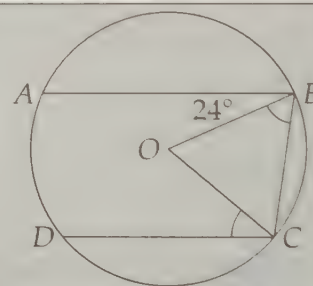
Problem 3.18: In the diagram, $XY = XZ = 8$ and $\angle X = 60^\circ$. Find YZ .



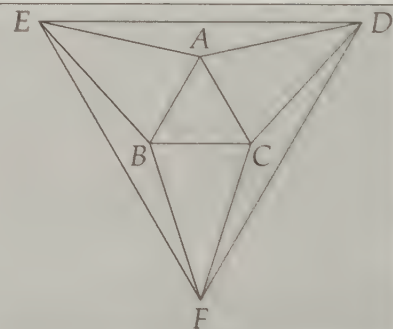
Problem 3.19: Find $\angle PBD$.



Problem 3.20: In the diagram, O is the center of the circle, $\overline{AB} \parallel \overline{CD}$, $\angle ABO = 24^\circ$, and $\angle OBC = \angle OCD$. Find $\angle BOC$.

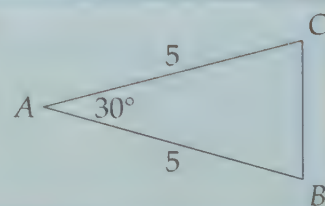


Problem 3.21: In the diagram at right, $AB = BC = AC$ and $AE = EB = BF = FC = CD = DA$ such that $\triangle ABC$ is completely inside $\triangle DEF$. Prove that $DE = EF = DF$.

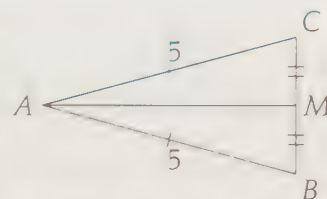


In this section, we investigate triangles that have two or three sides equal in length.

Problem 3.15: In the diagram, $AB = AC = 5$ and $\angle CAB = 30^\circ$. Find $\angle ACB$.



Solution for Problem 3.15: We're a little thin on information. We have a side equality that seems important. But we only have one triangle, so there's no obvious way to use the side equality. We might look for congruent triangles, but we'll have to somehow make two triangles. Splitting $\triangle ABC$ by connecting A to the midpoint, M , of BC gives us our congruent triangles.



Since $AB = AC$, $BM = CM$, and $AM = AM$, we have $\triangle AMB \cong \triangle AMC$ by SSS. This tells us that $\angle B = \angle C$, which is just enough to finish the problem. Since the angles of $\triangle ABC$ add up to 180° , we have

$$180^\circ = \angle A + \angle B + \angle C = 30^\circ + 2(\angle C).$$

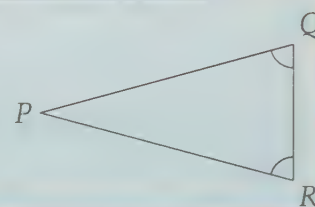
Solving for $\angle C$, we have $\angle C = (180^\circ - 30^\circ)/2 = 75^\circ$. $\square$

A triangle in which two sides are equal is called an **isosceles triangle**. The equal sides are sometimes called the **legs** of the triangle, and the other side the **base**.

Our general approach in Problem 3.15 can be used to show that if two sides of a triangle are equal, then the angles opposite those sides are equal. These two equal angles are often called the **base angles** of the triangle, and the other angle the **vertex angle**. As we saw in the last problem, if $\triangle ABC$ is isosceles with $\angle B = \angle C$, we have $\angle B = \angle C = (180^\circ - \angle A)/2$.

We might now wonder if this runs the other way: do equal angles imply equal sides?

Problem 3.16: Prove that if $\angle PQR = \angle PRQ$, then $PR = PQ$.



Solution for Problem 3.16: What's wrong with this solution?

Bogus Solution: We proceed as we did in Problem 3.15, by connecting P to M , the midpoint of $\overline{QR}$, as in Figure 3.6. Since $MR = MQ$, $\angle R = \angle Q$, and $PM = PM$, we have $\triangle PMR \cong \triangle PMQ$. Therefore, $PR = PQ$.

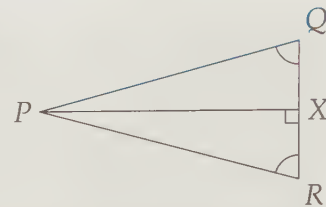


This failed solution uses 'SSA' Congruence, which we have shown in Section 3.5 doesn't work! We have to construct our triangles in a way that lets us use a valid congruence theorem.

Extra! The essence of mathematics is not to make simple things complicated, but to make complicated things simple.

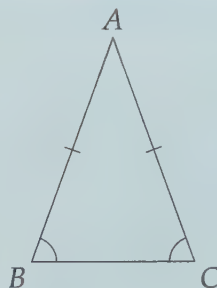
—S. Gudder

Instead of picking the point on $\overline{QR}$ that cuts it into two equal lengths, we make equal angles by connecting P to point X on $\overline{QR}$ such that $\overline{PX} \perp \overline{QR}$. Now we have $\triangle PXR \cong \triangle PXQ$ by AAS. Therefore, $PQ = PR$, as desired. (Note also that $QX = XR$, so that X is the same as point M in our Bogus Solution. This doesn't make that Bogus Solution any more correct, though!) $\square$



Putting our last two problems together gives us a pair of powerful tools.

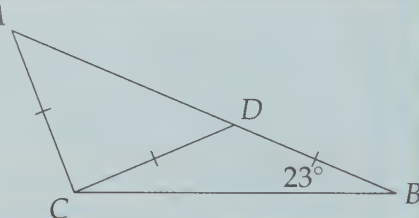
Important:



1. If $AB = AC$ in $\triangle ABC$, then $\angle B = \angle C$.
2. If $\angle B = \angle C$ in $\triangle ABC$, then $AB = AC$

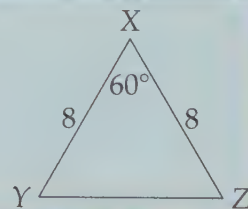
We'll try these tools on a couple more problems, and discover another important type of triangle.

Problem 3.17: In the diagram, $AC = CD = DB$, and $\angle B = 23^\circ$. Find $\angle A$.



Solution for Problem 3.17: Since $DC = DB$, we have $\angle DCB = \angle B = 23^\circ$. We could find $\angle CDB$ to get $\angle ADC$, or we can note that $\angle ADC$ is an exterior angle of $\triangle BCD$, so $\angle ADC = \angle DCB + \angle B = 46^\circ$. Since $AC = DC$, we have $\angle A = \angle ADC = 46^\circ$. $\square$

Problem 3.18: In the diagram, $XY = XZ = 8$ and $\angle X = 60^\circ$. Find YZ .



Solution for Problem 3.18: We can't do much about YZ right away, but we can find $\angle Y$ and $\angle Z$. Since $\angle X = 60^\circ$, $\angle Y + \angle Z = 180^\circ - 60^\circ = 120^\circ$. Since $XY = XZ$, we have $\angle Y = \angle Z$, so $\angle Y = \angle Z = 60^\circ$. Hence, all the angles of $\triangle XYZ$ are equal. Specifically, $\angle X = \angle Y$ means that $YZ = XZ$, so $YZ = 8$. $\square$

We call the triangle in Problem 3.18 an **equilateral triangle** because all of its sides are equal. As Exercises, you will use an approach a lot like the one above to show:

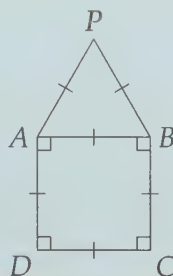
Important: If all three angles of a triangle are equal, then so are all three sides. Conversely, if all three sides are equal, then all three angles are 60° .



Therefore, in order to prove a triangle is equilateral, we can either prove all the sides are equal, or prove that all the angles are equal.

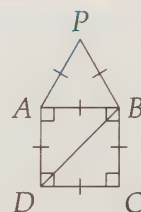
Let's put our isosceles and equilateral triangles to work.

Problem 3.19: Find $\angle PBD$.

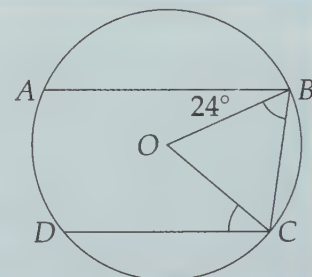


Solution for Problem 3.19: We start by drawing $\overline{BD}$ so we can see the angle we are after. Since $\triangle PAB$ is equilateral, $\angle PBA = 60^\circ$. $\triangle BAD$ is isosceles, since $AB = AD$. Since $\angle BAD = 90^\circ$, the other two angles of $\triangle BAD$ equal $(180^\circ - 90^\circ)/2 = 45^\circ$. Therefore, we have

$$\angle PBD = \angle PBA + \angle ABD = 105^\circ.$$



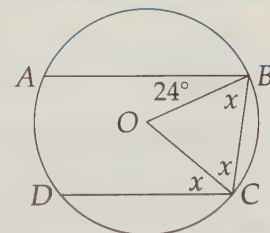
Problem 3.20: In the diagram, O is the center of the circle, $\overline{AB} \parallel \overline{CD}$, $\angle ABO = 24^\circ$, and $\angle OBC = \angle OCD$. Find $\angle BOC$.



Solution for Problem 3.20: We have parallel lines and an isosceles triangle ($OB = OC$ because they are radii of the same circle). So, we can do some angle-chasing. We let $\angle OBC = x$, so that we also have $\angle OCD = x$ from the given information and $\angle OCB = x$ from isosceles $\triangle OCB$. Since $\overline{AB} \parallel \overline{CD}$, we have $\angle ABC + \angle BCD = 180^\circ$. Therefore,

$$\angle ABO + \angle OBC + \angle BCO + \angle OCD = 180^\circ.$$

Substitution gives $24^\circ + 3x = 180^\circ$, so $x = 52^\circ$. Therefore, $\angle BOC = 180^\circ - 2x = 76^\circ$. $\square$



Extra! Mathematics is the gate and key to the sciences.

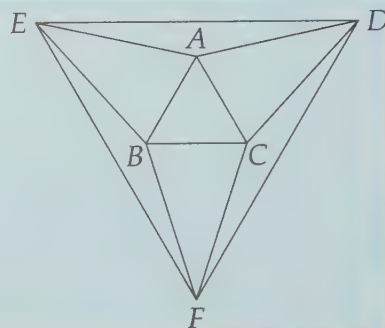


—Roger Bacon

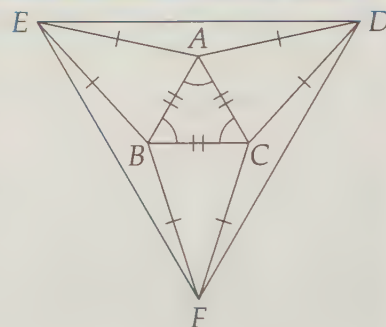
Concept: Assigning a variable to an angle will often help you with your angle-chasing. If you're stuck on an angle problem, assign one of the angle measures a variable and find other angles in terms of that variable. Hopefully, you'll eventually be able to build an equation you can use to solve for the variable.

We'll end this chapter with a proof involving equilateral and isosceles triangles.

Problem 3.21: In the diagram at right, $\triangle ABC$ is equilateral and $AE = EB = BF = CF = AD = CD$ such that $\triangle ABC$ is completely inside $\triangle DEF$. Prove that $\triangle DEF$ is equilateral.



Solution for Problem 3.21: In order to prove that $\triangle DEF$ is equilateral, we must show that either all its sides have the same length, or all its angles have the same measure. We start by marking all the given information. Our most useful tool in showing side lengths are equal is congruent triangles. Therefore, we look for triangles with $\overline{DE}$, $\overline{EF}$, and $\overline{FD}$ as corresponding sides. The triangles that stand out are $\triangle DEA$, $\triangle EFB$, and $\triangle FDC$. If we can show these are congruent, we'll have the desired $DE = EF = FD$.



In order to show that $\triangle DEA \cong \triangle EFB \cong \triangle FDC$, we need only show that $\angle DAE = \angle EBF = \angle FCD$, since we already know that the sides adjacent to these angles in our three triangles are all equal. To learn more about these angles, we look at the other angles at A , B , and C .

We already have $\angle CAB = \angle ABC = \angle BCA$. Furthermore, isosceles triangles $\triangle EAB$, $\triangle FBC$, and $\triangle DCA$ are all congruent by SSS, so their base angles are all equal. Now that we know three of the angles with vertex A equal the corresponding angles with vertex B , we can show $\angle DAE = \angle EBF$:

$$\angle DAE = 360^\circ - \angle EAB - \angle BAC - \angle DAC = 360^\circ - \angle FBC - \angle CBA - \angle EBA = \angle EBF.$$

Similarly, we have $\angle DAE = \angle FCD$. Therefore, by SAS we have $\triangle DEA \cong \triangle EFB \cong \triangle FDC$, which gives us the desired $DE = DF = EF$. So, $\triangle DEF$ is equilateral.

Try finding an alternate solution by showing that all the angles of $\triangle DEF$ are equal. $\square$

We typically prove a triangle equilateral by either proving all its side lengths are equal, or all its angles are equal. This often involves finding congruent triangles.

Concept: Mark the information you have in a problem on your diagram, particularly equal sides and equal angles. This will make congruent triangles easier to find.

We've explored isosceles and equilateral triangles, but we're not finished naming triangles yet! We have a name for two sides equal, and one for all three sides equal. If no two sides of a triangle are equal, the triangle is **scalene**.

Exercises

3.6.1 In $\triangle PQR$, $PQ = PR$ and $\angle P = 43^\circ$. Find $\angle Q$.

3.6.2 Prove that if $AB = AC$ in $\triangle ABC$, then $\angle ABC = \angle ACB$. (Note: You cannot simply state that the triangle is isosceles, so the base angles are equal. You are asked here to prove this fact.)

3.6.3

- Prove that if the three sides of a triangle are equal in length, then all three angles of the triangle have measure 60° .
- Prove that if the three angles of a triangle are equal, then all three sides of the triangle have the same length.

Note that for this problem, you are not allowed to state that the triangle is equilateral as your proof. You must prove the facts about equilateral triangles that you learned in the text.

3.6.4 Two angles of an equilateral triangle have measures $3x + 27^\circ$ and $2y - 4^\circ$. Find $x + y$.

3.6.5 Point O is the center of the circle in the diagram on the left below. Find $\angle AOB$ if $\angle OAB = 70^\circ$.

Hints: 79

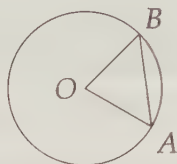


Figure 3.3: Diagram for Problem 3.6.5

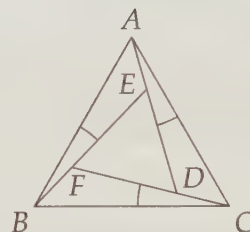
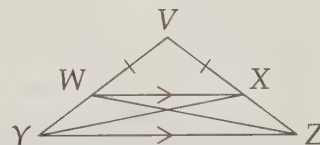


Figure 3.4: Diagram for Problem 3.6.6

3.6.6 Triangle ABC is equilateral in the diagram on the right above. Points D , E , and F are in triangle ABC such that $\angle CAD = \angle ABE = \angle BCF$. D lies on $\overline{CF}$, E lies on $\overline{AD}$, and F lies on $\overline{BE}$. Prove that triangle DEF is equilateral. **Hints:** 192, 574

3.6.7 In the diagram at right, $VW = VX$ and $\overline{WX} \parallel \overline{YZ}$.

- Prove that $WY = XZ$. **Hints:** 178, 518
- Prove that $YX = WZ$. **Hints:** 315



Extra! *I am interested in mathematics only as a creative art.*



—G. H. Hardy

3.7 Construction: Equilateral Triangle and Perpendicular Bisector

As a reminder, we once more repeat the rules of the game for constructions. The tools of classical construction problems are a compass and a straightedge. Notice that we don't say 'ruler.' You don't get to use your straightedge to measure – you can only draw lines. Similarly, you aren't allowed to use your protractor to measure or create angles.

The only operations you can perform with your compass and straightedge are:

1. Given a point, you can draw any line through the point.
2. Given two points, you can draw the line that passes through them both.
3. Given a point, you can draw any circle centered at that point.
4. Given a point and a segment, you can draw the circle with its center at that point and with radius equal in length to the length of the segment.
5. Given two points, you can draw the circle through one point such that the other point is the center of the circle.

Problems

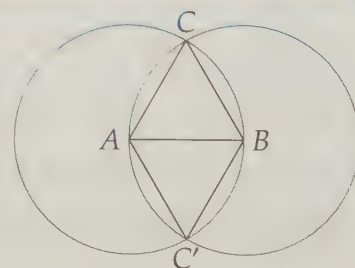
Problem 3.22: Construct an equilateral triangle with a compass and straightedge.

Problem 3.23: Draw a segment $\overline{AB}$. We call the line that is perpendicular to $\overline{AB}$ and passes through the midpoint of $\overline{AB}$ the **perpendicular bisector** of $\overline{AB}$. Construct the perpendicular bisector of $\overline{AB}$.

If you used your protractor to make a 60° angle to construct an equilateral triangle, go back to the beginning of this section and **READ THE RULES!**

Problem 3.22: Construct an equilateral triangle.

Solution for Problem 3.22: We start off by drawing a segment $\overline{AB}$. We want to find point C such that $AC = BC = AB$. Now we have a situation a lot like our SSS 'proof' of Problem 3.2. Since AC must equal AB , we know that C must be on the circle with center A and radius AB . We can construct this circle with our compass. Similarly, C is on the circle with center B and radius AB . Drawing this circle as well gives us the diagram to the right. Our two circles meet at C and C' . Both triangles ABC and ABC' are equilateral.



To prove that this construction works, we only have to prove that $AC = AB = BC$. Since each of these segments is a radius of a circle that is defined to have radius AB , clearly all three segments are equal. $\square$

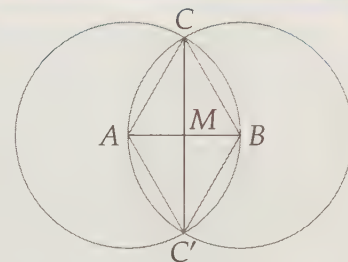
Important:

Construction problems are two-part problems. Find the construction, then prove that it works. The second part is usually pretty easy after you've found the construction, but don't forget to do it!

You'll see that in this book we often leave the proof part for the reader. Sometimes we'll ask you to supply the proof that our construction works as an Exercise.

Problem 3.23: Given a segment $\overline{AB}$, construct a line through the midpoint of $\overline{AB}$ that is perpendicular to $\overline{AB}$. We call this line the **perpendicular bisector** of $\overline{AB}$.

Solution for Problem 3.23: In our last construction, we found two points that appear to be directly above and directly below the midpoint of $\overline{AB}$. Perhaps connecting these points will give us the line we want. We start with the same construction we used to make an equilateral triangle. Then we draw $\overline{CC'}$, which meets $\overline{AB}$ at M as shown. By SSS, $\triangle CAC' \cong \triangle CBC'$, so $\angle ACC' = \angle BCC'$. Therefore, $\angle ACM = \angle BCM$. Since we also have $AC = BC$ and $\angle CAM = \angle CBM$ as we found in the previous problem, we have $\triangle ACM \cong \triangle BCM$ by ASA. So, $AM = BM$ and $\angle CMA = \angle CMB$. Since $\angle CMA + \angle CMB = 180^\circ$, we must have $\angle CMA = \angle CMB = 90^\circ$. Therefore, $\overleftrightarrow{CC'}$ is the perpendicular bisector of $\overline{AB}$. $\square$



Concept: Always be thinking about what you already know how to do when trying to do something new!



Exercises



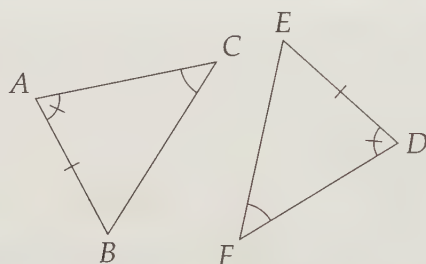
- 3.7.1 Given a segment $\overline{AB}$, divide the segment into 4 equal pieces with a straightedge and compass.
- 3.7.2 Is it necessary when constructing the perpendicular bisector of $\overline{AB}$ to use circles with radius AB ? Suppose we instead draw two intersecting circles with centers A and B and the same radius. Is the segment connecting the two points where these circles meet the perpendicular bisector of $\overline{AB}$?
- 3.7.3 Construct a 90° angle. **Hints:** 94
- 3.7.4 Construct a 30° angle. **Hints:** 504

3.8 Summary

Definition: Two figures are **congruent** if they are exactly the same – in other words, we can slide, spin, and/or flip one figure so that it is exactly on top of the other figure.

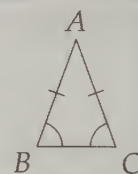
In this chapter we investigated the following theorems for proving that two triangles are congruent:

- **SSS Congruence:** If the sides of one triangle equal the sides of another triangle, then the triangles are congruent. (Section 3.2)
- **SAS Congruence:** If two sides of one triangle and the angle between them are equal to those of another triangle, then the two triangles are congruent. (Section 3.3)
- **ASA Congruence:** If two angles of one triangle and the side between them are equal to those of another triangle, then the two triangles are congruent. (Section 3.4)
- **AAS Congruence:** If two angles and a side of one triangle equal the corresponding angles and side in another triangle as shown below, then the triangles are congruent. (Section 3.4)



Important: Once we determine that two triangles are congruent, we can apply whatever we know about the sides and angles of one triangle to the other triangle. This obvious principle often goes by the fancy name '**Corresponding Parts of Congruent Triangles are Congruent**,' or **CPCTC**.

Definitions: A triangle in which two sides are equal is an **isosceles** triangle. The equal sides are the **legs** of the triangle and the other side is the **base**. The angle between the two equal sides is often called the **vertex angle** of the triangle, and the other two angles are the **base angles**, which are equal to each other.



Important: 1. If $AB = AC$ in $\triangle ABC$, then $\angle B = \angle C$.



2. If $\angle B = \angle C$ in $\triangle ABC$, then $AB = AC$

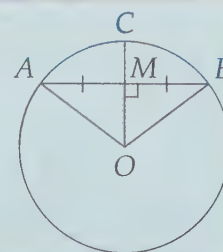
Definition: A triangle in which all three sides are equal is an **equilateral triangle**. All three angles of an equilateral triangle are 60° .

Important: If all three angles of a triangle are equal, then so are all three sides. Conversely, if all three sides of a triangle are equal, then all its angles equal 60° .

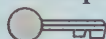
Throughout the book we will use congruent triangles to prove many important results. One of the first is:

Important:


If a radius of a circle bisects a chord of the circle that is not a diameter, then the radius must be perpendicular to the chord.



Problem Solving Strategies

Concepts:


- In more complicated geometry problems, mark side and angle equalities as you find them (particularly when you find non-obvious ones!)
- Dividing isosceles triangles in half by drawing a segment from the vertex between the equal sides to the midpoint of the base can be very effective.
- If you're stuck on an angle problem, assign one of the angle measures a variable and find other angles in terms of that variable. Hopefully, you'll eventually be able to build an equation you can use to solve for the variable.
- Mark the information you have in a problem on your diagram, particularly equal sides and equal angles. This will make congruent triangles particularly easy to find.
- Always be thinking about what you already know how to do when trying something new!

Things To Watch Out For!

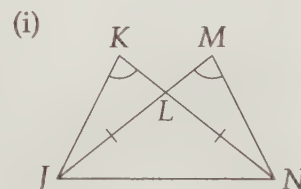
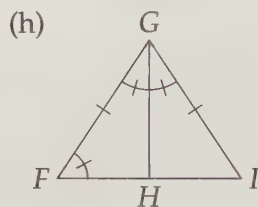
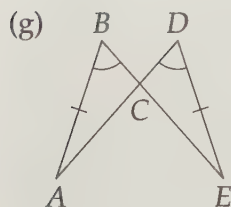
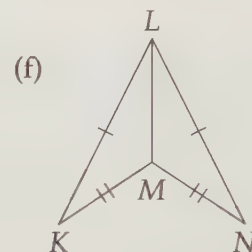
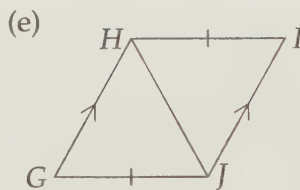
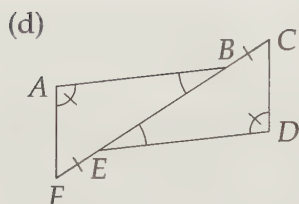
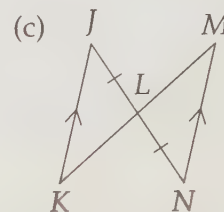
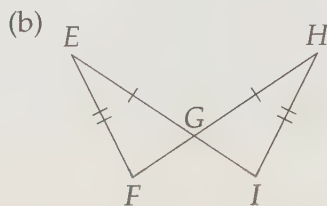
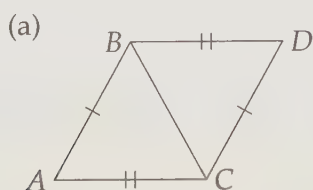
WARNING!!


SSA (Side-Side-Angle) is **not** a valid congruence theorem. If two sides of one triangle are equal to two sides of another, and the two triangles have equal corresponding angles *that are not the angles between the equal corresponding sides*, then the two triangles are **not necessarily congruent!**

REVIEW PROBLEMS

3.24 In each of the diagrams below, name all pairs of congruent triangles you can identify (without drawing more segments or naming more points). Write the relevant triangle congruences and explain

why the triangles are congruent. If there are not any pairs of triangles in a given diagram that must be congruent, state so.



3.25 Find the vertex angle of an isosceles triangle if one base angle is 6° less than half the vertex angle.

3.26 Find x in the diagram shown at left below.

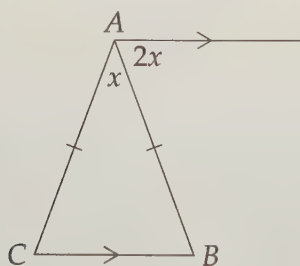


Figure 3.5: Diagram for Problem 3.26

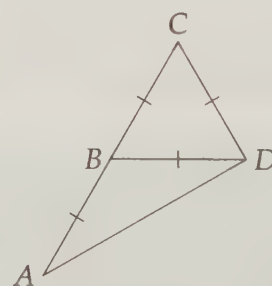


Figure 3.6: Diagram for Problem 3.27

3.27 Find $\angle A$ in the diagram at right above.

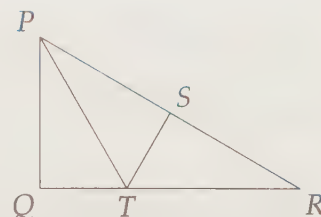
3.28 Segments $\overline{AB}$ and $\overline{CD}$ meet at point P such that $AP = CP$ and $BP = DP$.

(a) Is it always true that $\triangle APD \cong \triangle CPB$?

(b) Is it always true that $\overline{AD} \parallel \overline{BC}$? **Hints:** 547

3.29 Let PQR be a triangle as shown at right, and let S and T be points on sides $\overline{PR}$ and $\overline{QR}$, respectively. Triangles PTS , PTQ , and RTS are congruent.

- Show that $\angle PST = 90^\circ$.
- Find the measures of the angles of the triangle PQR .



3.30 In triangle NOK at left below, $KO = KN$. Let I be the point on $\overline{KN}$ such that $OI = ON$. Let A be the point on $\overline{NO}$ such that $IA = IN$.

- Find three angles in the diagram equal in measure to $\angle N$.
- Prove that $\overline{IA} \parallel \overline{KO}$.

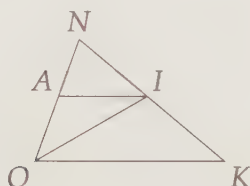


Figure 3.7: Diagram for Problem 3.30

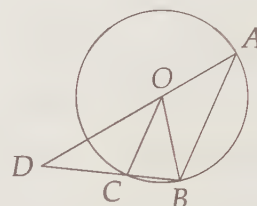


Figure 3.8: Diagram for Problem 3.31

3.31 In the diagram at right above, O is the center of the circle, $OA = CD$, and $\angle AOB = \angle OCD$.

- Show that $\triangle BOA \cong \triangle OCD$.
- Prove that $\overline{OC} \parallel \overline{AB}$.
- Show that $\angle OCB = 2\angle OBA$.
- Find $\angle OBA$.

3.32 In the diagram at left below, $AD = BD = AE$ and $DE = EC$. Prove that $AC = BE$. **Hints:** 71

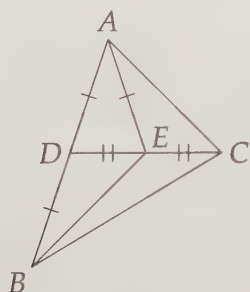


Figure 3.9: Diagram for Problem 3.32

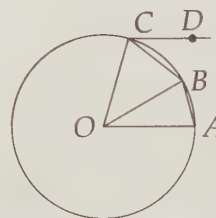


Figure 3.10: Diagram for Problem 3.33

3.33 In the figure at right above, $\angle BOC = 42^\circ$, O is the center of the circle, $\overline{CD} \parallel \overline{OA}$, and $\angle BAO = 2\angle BCD$.

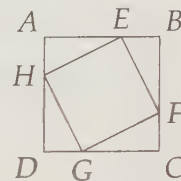
- Find $\angle CBO$.
- Show that $\angle BOA = 180^\circ - 4\angle BCD$.
- Find $\angle BOA$. **Hints:** 145

CHAPTER 3. CONGRUENT TRIANGLES

3.34 $\overline{AB}$ and $\overline{XY}$ are diameters of $\odot O$. Prove that $AX = BY$.

3.35 In triangle ABC , let M be the midpoint of $\overline{BC}$. Prove that if $MA = MB = MC$, then $\angle BAC = 90^\circ$.
Hints: 546

3.36 All four angles of a square are 90° and all four sides have the same length. Square $EFGH$ is inscribed in square $ABCD$ as shown at right, meaning that each vertex of $EFGH$ is on a side of $ABCD$.



- Show that $\angle AHE = \angle BEF$.
- Prove that triangles AEH , BFE , CGF , and DHG are congruent.

3.37 In the figure at left below, $\triangle ABC$ is equilateral, $AE = DC = BF$, and $EB = CF = AD$, such that $\triangle ABC$ is completely inside $\triangle DEF$.

- Prove that $\triangle DEF$ is equilateral. Hints: 73
- Would $\triangle DEF$ still be equilateral if $\triangle ABC$ were not fully inside $\triangle DEF$?

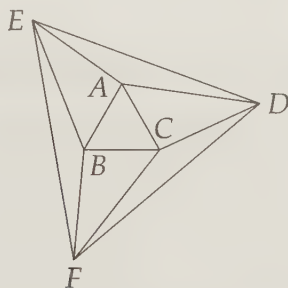


Figure 3.11: Diagram for Problem 3.37

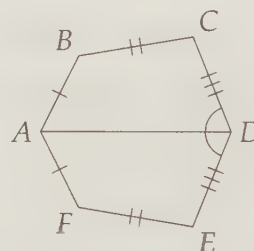
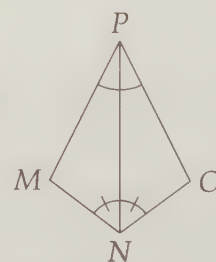


Figure 3.12: Diagram for Problem 3.38

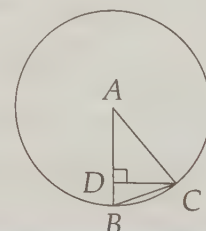
3.38 In $ABCDEF$ at right above, $AB = FA$, $BC = EF$, $CD = DE$, and $\angle CDA = \angle ADE$. Prove that $\angle ABC = \angle EFA$. Hints: 33, 500

3.39 $\overline{NP}$ in the diagram bisects both $\angle MNO$ and $\angle MPO$. Which of the following statements must be true? For each statement that must be true, prove that it must be true. For each statement that may not be true, explain why it may not be true.



- $NO = MN$.
- $NO = OP$.
- $\overline{MP} \parallel \overline{NO}$.
- ★ $\overline{MO} \perp \overline{NP}$.

3.40 In the diagram at right, A is the center of the circle and $\overline{CD} \perp \overline{AB}$. We are given that $\angle ACD = 2\angle DCB$.



- Let $\angle DCB = x$. Find expressions for $\angle ABC$ and $\angle BAC$ in terms of x .
- Find $\angle ABC$.

Challenge Problems

3.41 The measures of two angles of an isosceles triangle are $3x + 4^\circ$ and $x + 17^\circ$. Find all possible values of x . **Hints:** 148

3.42 Triangle ABC is equilateral in the diagram at left below, and $ABDE$, $BCFG$, and $CAHI$ are squares. Prove that triangle DFH is equilateral. **Hints:** 274, 534

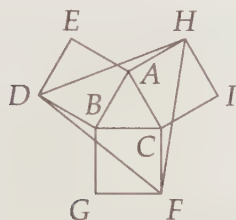


Figure 3.13: Diagram for Problem 3.42

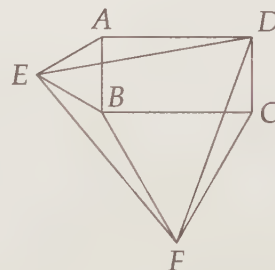


Figure 3.14: Diagram for Problem 3.43

3.43 Let $ABCD$ be a rectangle, meaning that all four of its angles are right angles. Construct equilateral triangles ABE and BCF externally on sides $\overline{AB}$ and $\overline{BC}$ as shown at right above. Prove that triangle DEF is equilateral. **Hints:** 88, 482, 582

3.44 $\triangle ABC$ has a right angle at $\angle C$. Points D and E are on $\overline{AB}$ as shown at left below such that $AD = AC$ and $BE = BC$. Find $\angle DCE$. **Hints:** 401, 232

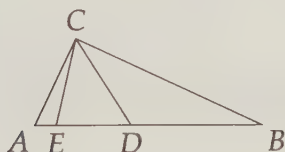


Figure 3.15: Diagram for Problem 3.44

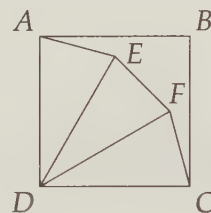


Figure 3.16: Diagram for Problem 3.45

3.45★ Let $ABCD$ be a square, and let E, F be points such that $DA = DE = DF = DC$ and $\angle ADE = \angle EDF = \angle FDC$. Prove that triangle BEF is equilateral. **Hints:** 280, 86, 575

3.46 In triangle ABC in the diagram below, D and E are points on side $\overline{AB}$, and F and G are points on side $\overline{AC}$, such that $AD = DG = GB = BC = CE = EF = FA$. Find $\angle BAC$. **Hints:** 436

